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**Java SE 27 & JDK 27**

# Interface TemporalQuery<R>

**Type Parameters:**

R - the type returned from the query

**Functional Interface:**

This is a functional interface and can therefore be used as the assignment target for a lambda expression or method reference.

---

```
@FunctionalInterface
public interface TemporalQuery<R>
```

Strategy for querying a temporal object.

Queries are a key tool for extracting information from temporal objects. They exist to externalize the process of querying, permitting different approaches, as per the strategy design pattern. Examples might be a query that checks if the date is the day before February 29th in a leap year, or calculates the number of days to your next birthday.

The TemporalField interface provides another mechanism for querying temporal objects. That interface is limited to returning a long. By contrast, queries can return any type.

There are two equivalent ways of using a TemporalQuery. The first is to invoke the method on this interface directly. The second is to use TemporalAccessor.query(TemporalQuery):

```
// these two lines are equivalent, but the second approach is recommended
result = thisQuery.queryFrom(temporal);
result = temporal.query(thisQuery);
```

It is recommended to use the second approach, query(thisQuery), as it is a lot clearer to read in code.

The most common implementations are method references, such as LocalDate::from and ZoneId::from. Additional common queries are provided as static methods in TemporalQueries.

**Implementation Requirements:**

This interface places no restrictions on the mutability of implementations, however immutability is strongly recommended.

**Since:**

1.8

## Method Summary

| All Methods | Instance Methods | |
|---|---|---|

| Modifier and Type | Method | Description |
|---|---|---|
| R | **queryFrom**(TemporalAccessor tempor | Queries the specified temporal object. |

## Method Details

### queryFrom

`R queryFrom(TemporalAccessor temporal)`

Queries the specified temporal object.

This queries the specified temporal object to return an object using the logic encapsulated in the implementing class. Examples might be a query that checks if the date is the day before February 29th in a leap year, or calculates the number of days to your next birthday.

There are two equivalent ways of using this method. The first is to invoke this method directly. The second is to use `TemporalAccessor.query(TemporalQuery)`:

```
// these two lines are equivalent, but the second approach is recommended
result = thisQuery.queryFrom(temporal);
result = temporal.query(thisQuery);
```

It is recommended to use the second approach, `query(thisQuery)`, as it is a lot clearer to read in code.

**Implementation Requirements:**

The implementation must take the input object and query it. The implementation defines the logic of the query and is responsible for documenting that logic. It may use any method on `TemporalAccessor` to determine the result. The input object must not be altered.

The input temporal object may be in a calendar system other than ISO. Implementations may choose to document compatibility with other calendar systems, or reject non-ISO temporal objects by `querying the chronology`.

This method may be called from multiple threads in parallel. It must be thread-safe when invoked.

**Parameters:**

`temporal` - the temporal object to query, not null

**Returns:**

the queried value, may return null to indicate not found

**Throws:**

`DateTimeException` - if unable to query

`ArithmeticException` - if numeric overflow occurs

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Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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